

HAOWEI ZHANG

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📌 Research Interests: Multimodal Reasoning, Video Understanding & Generation, Efficient Inference

🎓 EDUCATION

Fudan University, Shanghai, China 2025 – Present

Ph.D. student in FudanNLP, supervised by Prof. Xipeng Qiu and Jinlan Fu

Technical University of Munich (TUM), Munich, Germany 2021 – 2024

M.Sc. in Robotics, Cognition, Intelligence

Final Grade: 1.7 / 1.0 (German system, equivalent to 91 / 100)

Tongji University, Shanghai, China 2017 – 2021

B.Eng. in Mechatronics

Final Grade: 4.54 / 5.0 (equivalent to 91 / 100)

👤 EXPERIENCE

MOSI Shanghai, China 2026.03 –

🔗 <https://mosi.cn/>

Research Intern Leader: Qinyuan Cheng, Shimin Li, Prof. Xipeng Qiu

- fine-grained video generation

Yale NLP Lab remote 2024.10 – 2025.02

🔗 <https://yale-nlp.github.io/>

Research Assistant Leader: Yilun Zhao, Prof. Arman Cohan

- video understanding and reasoning of multimodal LLMs
- one paper accepted at CVPR as co-first author

LMU Research Group Munich, Germany 2023.11 – 2024.03

🔗 <https://tresp-lab.github.io/>

Research Assistant Leader: Zhen Han, Jindong Gu, Prof. Volker Tresp

- improving reasoning and calibration capabilities of multimodal LLMs
- one paper accepted at EMNLP 2024 Findings as first author

Yale NLP Lab remote 2023.03 – 2023.10

🔗 <https://yale-nlp.github.io/>

Research Assistant Leader: Yilun Zhao, Prof. Arman Cohan

- table-to-text generation capabilities of LLMs
- one paper accepted at EMNLP 2023 Industry Track as co-first author
- one paper accepted at EMNLP 2024 Demo Track as first author

🔍 PROJECTS

HERMES: KV Cache as Hierarchical Memory for Efficient Streaming Video Understanding

2025.08 – 2025.12

🔗 Paper Link

- proposing a training-free architecture designed for real-time and accurate understanding of video streams by reusing hierarchical KV cache
- under review as first author

GroundedPRM: Tree-Guided and Fidelity-Aware Process Reward Modeling for Step-Level Reasoning

2024.12 – 2025.03

🔗 Paper Link

- proposing a tree-guided and fidelity-aware framework for automatic process supervision
- under review as second author

MMVU: A Comprehensive Benchmark for Expert-Level Multi-Disciplinary Video Understanding and Reasoning

2024.10 – 2025.02

🔗 Paper Link | 🔗 Project Link

- comprehensive expert-level, multi-disciplinary benchmark for evaluating foundation models in video understanding
- attracting great attention from the community (download >27k on HF), used by Qwen3-VL, Kimi-VL, Seed2.0 testing, etc.
- accepted at CVPR 2025 as co-first author

Visual Question Decomposition on Multimodal Large Language Models

2023.10 – 2024.03

🔗 Paper Link | 🔗 Project Link

- development of a novel evaluation framework to assess the quality of generated sub-questions
- proposing a finetuning pipeline to enhance the capabilities of multimodal LLMs for visual question decomposition
- instructing multimodal LLMs to perform selective question decomposition
- accepted at EMNLP 2024 Findings as first author

OpenT2T: An Open-Source Toolkit for Table-to-Text Generation

2023.07 – 2023.09

🔗 Paper Link | 🔗 Project Link

- first open-source toolkit for table-to-text generation, designed to reproduce performance comparison and to expedite the development of new models
- accepted at EMNLP 2024 Demo Track as first author

Investigating Table-to-Text Generation Capabilities of Large Language Models in Real-World Information Seeking Scenarios

2023.03 – 2023.07

🔗 Paper Link | 🔗 Project Link

- investigation of the table-to-text capabilities of different LLMs within real-world information seeking scenarios
- demonstrating that high-performing LLMs can effectively serve as table-to-text generators, evaluators, and feedback generators
- accepted at EMNLP 2023 Industry Track as co-first author

💖 COMMUNITY SERVICE

Organizer of Knowledge-Intensive Multimodal Reasoning @ ICCV 2025

Reviewer of Reasoning and Planning for LLMs @ ICLR 2025

Reviewer of ACL Rolling Review 2025

🏆 HONORS AND AWARDS

Outstanding Graduate of Tongji University

2021.04

DAAD-Scholarship (German national level scholarship)

2020.11

⚙️ SKILLS

- Research: Python / Natural Language Processing / (Multimodal) Large Language Models
- Languages: English, German - Fluent / Chinese - Native speaker